

Digital Voice Recorder

Clay Nimmo

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Executive Summary

An analogue to digital converter's input conditioning signal has been successfully constructed for the following design constraints:

- 5V input.
- Designed within one TL974 operational amplifier.
- Signal centred at 2.5V
- Passband of at least 3.4kHz.
- Attenuate by 256x the passband voltage by 7.8kHz (-48dB)
- Appropriate gain to maximize the signal within the 5V ceiling.

The circuit has successfully obtained the input signal from the microphone by removing the DC bias, attenuating with an appropriate anti-aliasing filter, and amplifying the signal to remain audible. This culminated into an observed digital recording that maintained clarity, with similar audio quality to a radio.

While the overall circuit's characteristics vary significantly through the tolerances, the worst-case tolerance examples are still within the design constraints, and so is sufficient for large scale production.

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1 Introduction

This report highlights the design of the input signal conditioning for a microphone to use in a digital voice recorder. The design is split into three sections: the microphone interfacing and DC removal, anti-aliasing filter, and the gain. The report will step through the design decisions and the underlying theory for why the choices made are optimal. This involves component calculations, circuit simulations, and experimental results. To understand each section, a brief description is given, expanded in more detail further in the report:

The DC offset is required to capture as much of the signal as possible, by centring the DC voltage between the minimum and maximum, giving the largest possible amplitude of the output waveform. This DC removal takes form as a low cutoff frequency high pass filter. Since, as per the datasheet, the microphone only registers frequencies above 100Hz, the high pass only removes the low frequency DC part of the circuit while not attenuating the microphone's output.

The anti-aliasing filter is required as part of the voice recorder. To avoid distortion of the signal from aliasing, the signal must be fully attenuated at half the sampling frequency. In the context of the ATmega microcontroller, the sampling rate is 15.625kHz, and so must be fully attenuated by 7.81kHz. Since the microcontroller samples at 8 bits, the signal is considered fully attenuated when the magnitude is 256 times smaller than the passband gain such that it can no longer be encoded, corresponding to -48db. To keep as much of voice signal as possible, the difficulty of designing the filter is in keeping the passband as large as possible.

The gain is required to give a cleaner signal, where the peak-to-peak voltage is more pronounced. This allows cleaner sampling, improving the audio quality, while increasing the volume of the recording. For the highest quality, the maximum voltage must be as close to the 5V ceiling as possible. However, it is important to give the signal a safe margin below 5V, to prevent the risk of clipping. To achieve this, a gain between 20-30db is optimal, since the microphone gives a V_{pp} near 100mV.

2 Background and Literature Review

Aliasing is a phenomenon in signal processing where there are not enough samples to accurately reconstruct the waveform, reducing the accuracy and quality of the signal. According to the Nyquist Sampling Theorem, the signal can only be perfectly reconstructed if the sampling rate is double the maximum frequency in the signal [1]. To resolve this, the sampling rate can either be increased, or an anti-aliasing filter can be implemented. Since the microcontroller in use (ATmega) has a fixed sampling rate, the anti-aliasing filter is the only viable option.

For the anti-aliasing filter, there are two circuit topologies used for this design, being the second order Sallen Key RC low pass topology, and the low pass inverting amplifier circuit. These topologies make use of an op amp, simplifying the filter design as the transfer function of each stage can simply be multiplied together, without additional loading. Both topologies also allow some gain, but to avoid overloading the op amp, the voltage differential between the terminals need to be low. Because of this, the gain is left to the last stages. Within the inverting amplifier circuit, the gain does not impact the cutoff frequency since the low pass only has one real pole as opposed to the Sallen Key's complex conjugate poles. For a steeper cutoff, the Chebyshev model is used for component selection, increasing the passband in exchange for a ripple. The passband ripple attenuates and boosts some frequencies, causing an unbalanced equalization and distorting the output signal [2]. To balance the distortion and cutoff, a maximum passband gain of 1.5db is set while calculating the filter specification.

For the recorder to be effective, it needs to capture the key frequencies of human speech. In quantized audio transmissions, which is relevant to the frequency processing for the voice recorder, the international standard bandwidth allocated to maintain vocal clarity is between 300Hz and 3400Hz [3]. The majority of syllable clarity is between 750Hz and 2850Hz which contributes to ~83% articulation, where 2850 to 7000 only increases articulation by approximately 15% [4]. Therefore, while 3400Hz is ideal, 2850Hz is the absolute minimum to maintain a clear voice recording. Following the Nyquist rate, this means the required sampling rate is 6.8kHz, which is beneath the 7.8kHz limit and so is achievable with an appropriate anti-aliasing filter.

For the op amps that are used for the gain, one terminal must be supplied with a reference voltage of 2.5V rather than being tied to ground, to centre the signal. So, in the case of the Sallen Key topology, an additional op amp is required to buffer the reference voltage, while it is not necessary for the inverting amplifier circuit since the terminals of op amps draw no current.

3 Circuit Design

3.0 Design Specifications

The circuit is designed for the following specifications:

- 5V input voltage.
- Minimize cost by using one integrated circuit (IC), allowing only four op amps.
- -48db attenuation at 7.8kHz.
- 20-28db passband gain, transforming 100mV microphone input to as close to 2.5vpp as possible.
- Designed for 100mVpp input from the microphone.
- Op amp common mode input voltage between 1.15V and 3.85V, to avoid clipping

3.1 Microphone Interfacing

The microphone connection is established from two parts: a dropdown resistor to lower the input voltage bias, and a high pass filter buffered using an op amp with a small amplification. The high pass filter further removes the DC component, and buffers the microphone input from the rest of the circuit. The dropdown resistor is derived from the microphone's unique bias current. This current was measure experimentally using an ammeter, which measured at 0.2mA. So, using ohms law, the required pulldown resistor is 12.5k, giving the closest E12 series resistor of 12k.

The high pass filter is designed with a cutoff frequency of 5Hz, such that the signal will fully pass by 100Hz from the 20db per decade slope. Using standard values, the capacitor is set to 3.3uF, and so following the standard high pass filter formula, requires an impedance of 10k. While the 10k resistor is part of the gain circuit, the second resistor only impacts the gain of the stage and has no impact on the pole, much like with the low pass variant. To offset the output voltage, the positive terminal must use a reference voltage of 2.5V, achieved using simple voltage divider of equal resistor values. The circuit designs for the microphone interfacing are shown in Figure 1 and Figure 2.

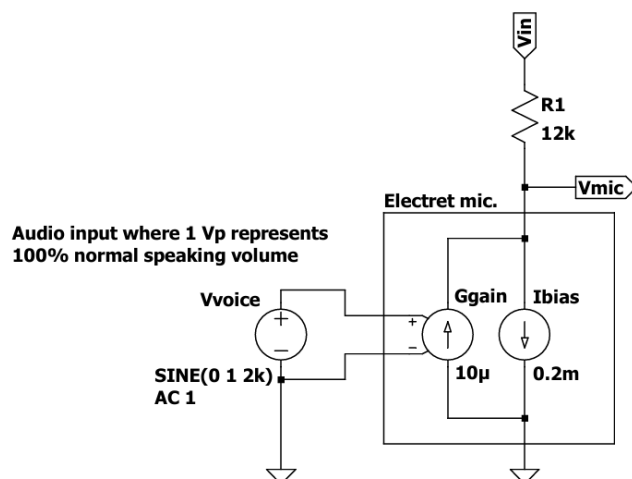


Figure 1 LTSpice circuit design depicting the microphone connection, showing a pulldown resistor of 12k, and a bias current of 0.2mA

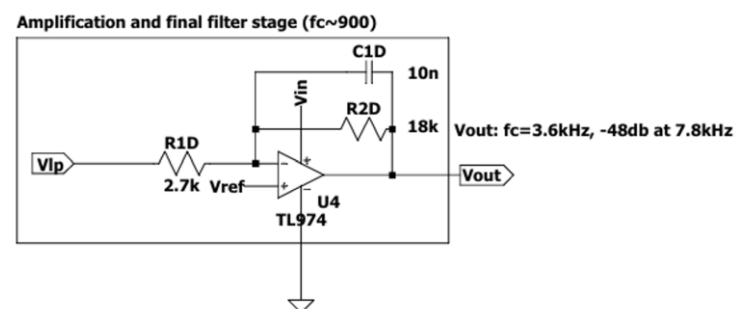


Figure 2 LTSpice circuit design depicting the 5Hz high pass filter buffer stage for the microphone connection. Vtest is a test voltage source of 2.5V offset, 100mV amplitude. Vtest is used instead of Vmic for more consistent plots.

3.2 Anti-Aliasing Filter

The anti-aliasing filter is designed in MATLAB, using the `cheb1` and `cheby1ord` functions to get the required transfer function for a circuit with a cutoff frequency of 3.6kHz, and reaches -48dB by 7.8kHz. Then, the poles of the transfer function are extracted to match coefficients to obtain the component values. The Chebyshev filter type is chosen, since it can reach the desired cutoff with a lower order filter design, helping to keep the design within one IC.

From the specifications, MATLAB returned a required filter order of five, meaning the transfer function has two complex conjugate poles, and one real pole. This section will only demonstrate the fourth order stages from two cascading unity gain Sallen key filters, since the final stage can be combined with the gain. Unity gain filters are chosen, since the version with the gain requires wasting one op amp for a voltage reference, and so would require another IC since the circuit needs two for the fourth order stage, one for the microphone buffer, and one for the gain.

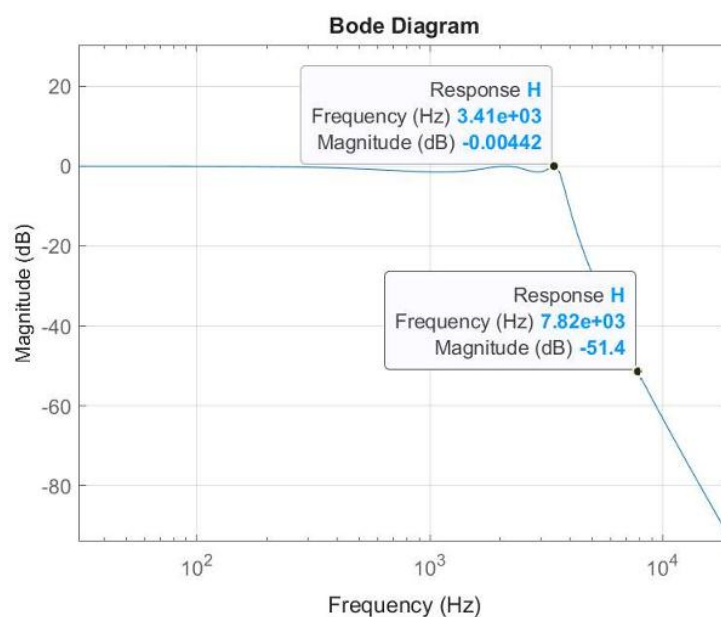


Figure 3 Fifth order Chebyshev filter bode plot, generated from MATLAB.

Figure 3 shows the frequency response of the MATLAB generated transfer function. The labels show that the transfer function has a passband of 3.4kHz, and drops beyond -48dB at 7.8kHz. Therefore, this transfer function meets the specifications, and so it can move onto component selection.

```
[z, p, k] = tf2zpk(b, a); %get the zeros, poles, and gain of the transfer function
```

```
%%%%%% stage A %%%%%%%%%%
```

```
a1A = -p(1)-p(2);
```

```
a2A = p(1)*p(2);
```

```
wnA = sqrt(a2A);
```

```
QA = wnA/a1A;
```

```
KA = 1;
```

```
nA = (2*QA)^2;
```

```
C1A = 3.3e-9;
```

```
C2A = nA*C1A;
```

```
R1A = 1/(wnA * C1A * sqrt(nA));
```

```
R2A = R1A;
```

Listing 1 MATLAB code to get the component values of the first second order unity gain Sallen Key topology, using the transfer function poles.

Listing 1 demonstrates the code to calculate the component values of a second order unity gain low pass Sallen Key filter. It extracts the first complex conjugate poles obtained from the transfer function using the tf2zpk function, which is used to obtain the cutoff frequency (wnA) and the gain (QA). These two variables are then used to solve for the component values using the transfer function of the topology, by setting the gain to one (for unity gain) and fixing one capacitor value to lower the number of unknowns. The 3.3nF capacitor value was chosen since it is a common standard value that is not too small or large. The second filter stage uses identical code, differing only in using the poles at indexes 3 and 4 instead. The output of the MATLAB code is displayed in Table 1, where Fc is $\omega_n/2\pi$.

Stage 1	Stage 2
Fc = 3.5kHz	Fc = 2.3kHz
R1 = 1060Ω	R1 = 6.6Ω
R2 = 1060Ω	R2 = 6.6 Ω
C1 = 3.3nF	C1 = 3.3nF
C2 = 544nF	C2 = 33.4nF

Table 1 Calculated component values for the two second order Sallen Key stages.

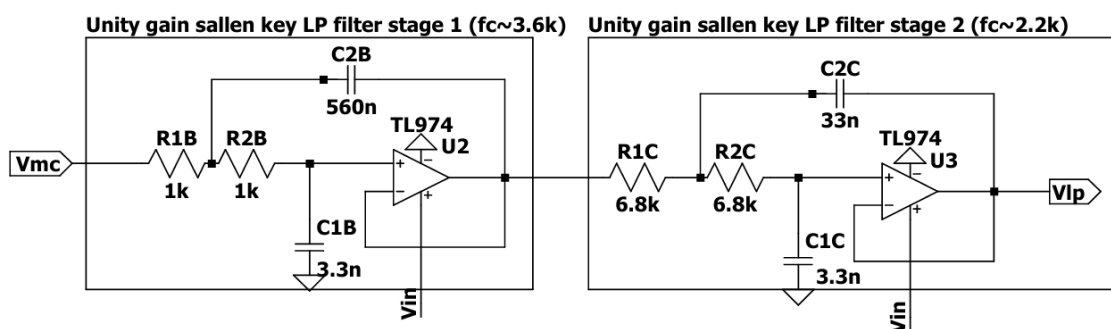


Figure 4 Second order unity gain low pass filters, cascaded to make a fourth order transfer function.

The component values in Table 1 are replaced with the closest E12 series components, and placed in the circuit diagram depicted in Figure 4. The slight differences in component values resulted in a

$\pm 100\text{Hz}$ different to the cutoff frequency. However, since the original transfer function overshoot the -48dB requirement, this slight difference will not make the filter less effective.

3.3 Amplification and Final Filter Stage

The final circuit stage uses the low pass inverting amplifier model, where it is similar to the first stage but the capacitor is instead parallel to the negative feedback resistor. From the transfer function, the cutoff frequency is determined only by the two negative feedback components, while the resistor at the input only couples with the second resistor to give the gain [5]. As a first order function, it's contribution can be extracted in MATLAB by taking the only real pole. Since it has a different transfer function, the code in Listing 1 cannot be reused.

```
wnC = abs(real(p(5))); % get the real pole

C1P = 10e-9; % set the capacitor to 10nF

R2P = 1 / (wnC * C1P);
KC = -4; % set the desired gain
R1P = -R2P / KC;

KC = -R2P/R1P;
```

Listing 2 MATLAB code to calculate the component values of the final filter stage

The code output the following component values: $R1=4462\Omega$, $R2=17.8\text{k}\Omega$, $C1=10\text{nF}$, with a cutoff frequency of 900Hz . To get the required total gain, including from the first stage, the value of $R1$ was manually decremented from the E12 standard values in LTSpice until the simulation output using the test voltage returned a passband gain of 28dB . As defined by the transfer function, modifying $R1$ does not impact the frequency response of the circuit given $R2$ is constant. The resistor value found to give this result is 2.7k . Much like the first stage, this stage's positive terminal also requires a reference voltage. However, since the input terminals ideally draw no current, the same reference voltage can be used from the first stage without loading. The circuit design for this stage is shown in Figure 5.

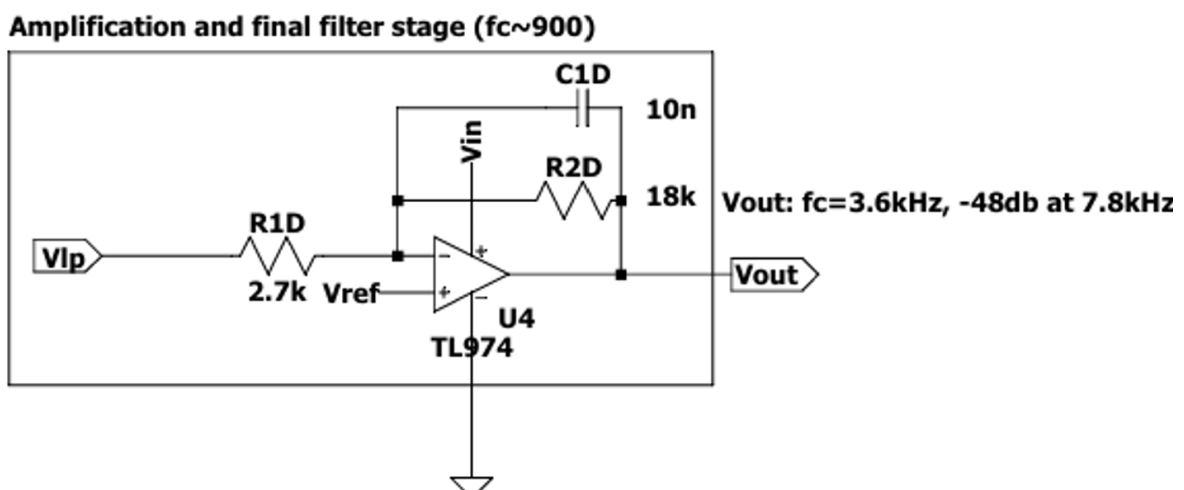


Figure 5 LTSpice circuit design of the final low pass and gain stage.

3.4 Full Circuit

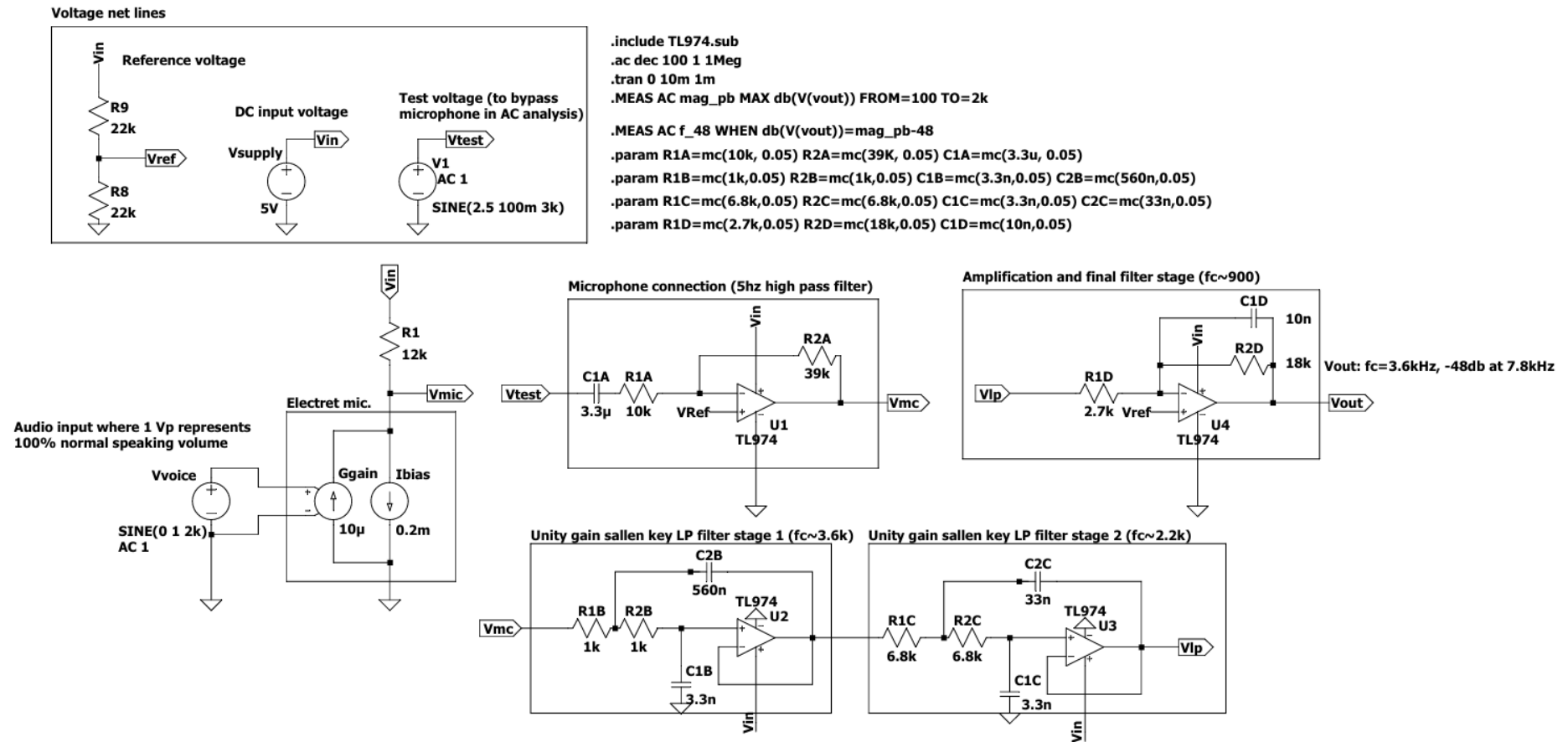


Figure 6 Full circuit design.

To help interpreting the flow of the labelled nets, the stages go from: electret mic or V_{test} -> microphone connection -> sallen key LP filter stage -> final LP stage and gain -> V_{out} . The component values are parameterized, but their labels have been reverted to the standard value for better viewing.

4 Simulations in LTspice

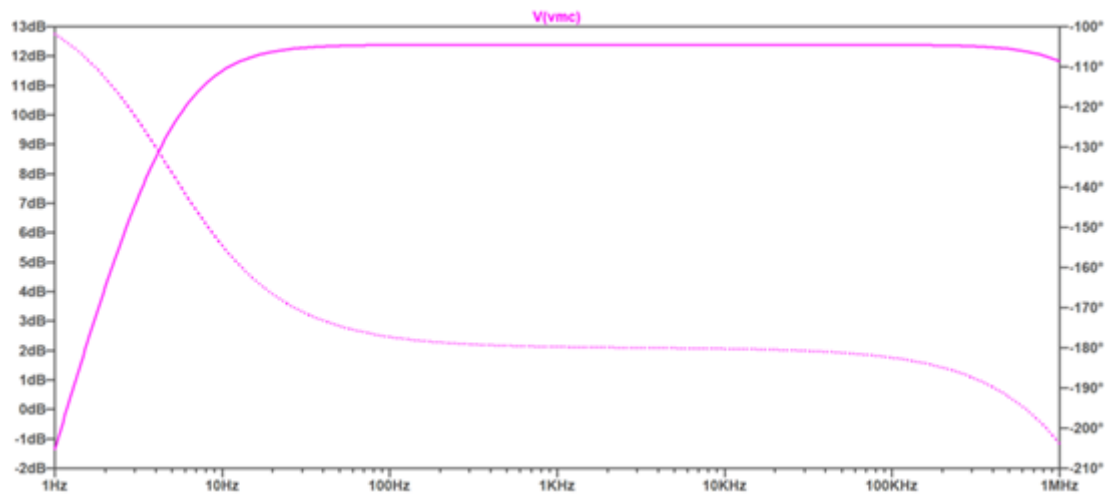


Figure 7 Bode plot of the output to the high pass buffer stage, showing that the signal is fully active before 100Hz, and a gain of 12db

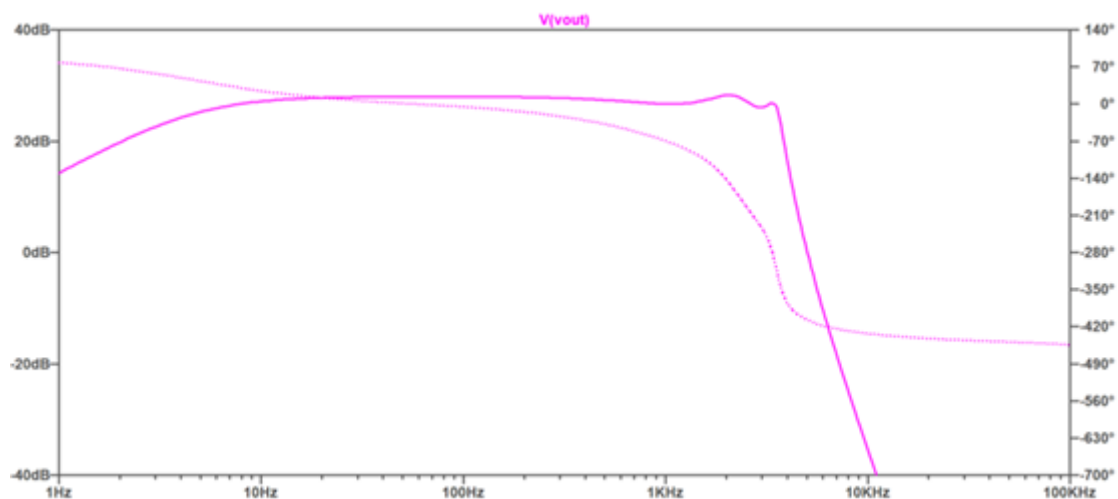


Figure 8 Ideal frequency response output of the circuit

Figure 7 and Figure 8 demonstrate the ideal frequency responses of the circuit, where the former is measured after the first stage while the latter is the output of the full circuit. Figure 7 demonstrates the high pass filter function, where low frequency signals are appropriately attenuated, while the signal is fully active by 100Hz. Figure 8 showcases the overall transfer function, where the passband is flat until it has a slight ripple around 3kHz, before having a steep drop-off at 3.6kHz, differing from the MATLAB code by 200Hz. The signal shows an average gain of 28dB, where it attenuates to -20dB at 7.8kHz. Excluding the high pass and the gain, the bode plot matches the MATLAB plot in Figure 3, demonstrating that the implemented transfer functions are accurate to the circuit in theory. However, for a better representation of the real-world design, the component tolerances are simulated.

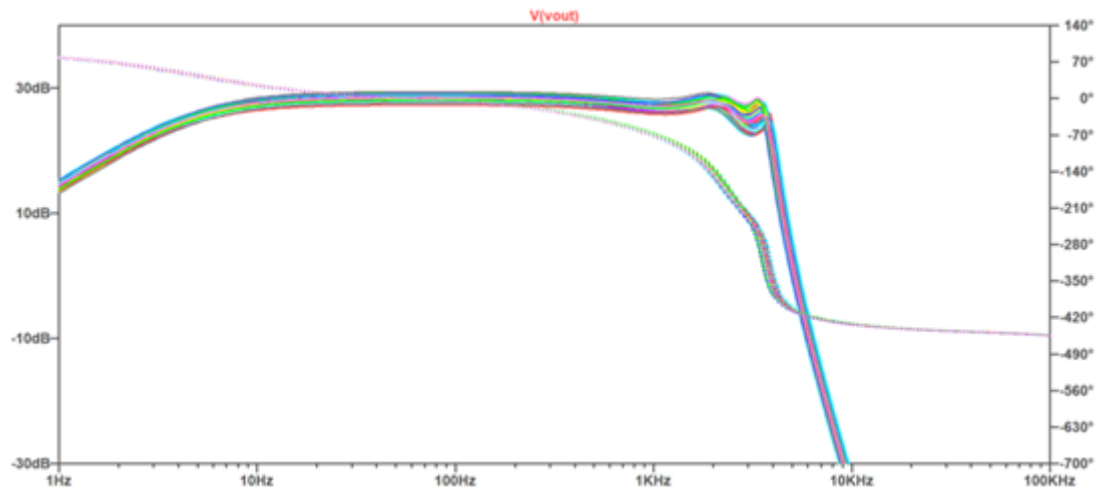


Figure 9 Parameterized output for 100 random component values within the tolerance.

Figure 9 showcases the bode plot of the entire circuit for 100 random component values taken with a 5% tolerance. It shows that the gain ranges from 26dB to 30dB, while the -48dB cutoff is reached from 6kHz to 8kHz. So, while there are some component values that will not reach the requirement, this requirement failure is only within less than 5% of the component values, and so is sufficiently reliable.

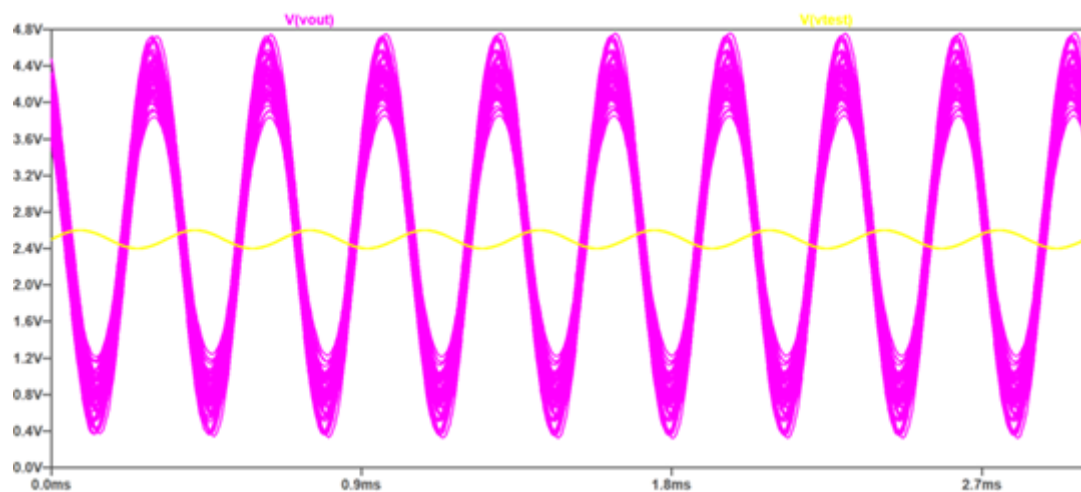


Figure 10 Transient response of the circuit at 3kHz

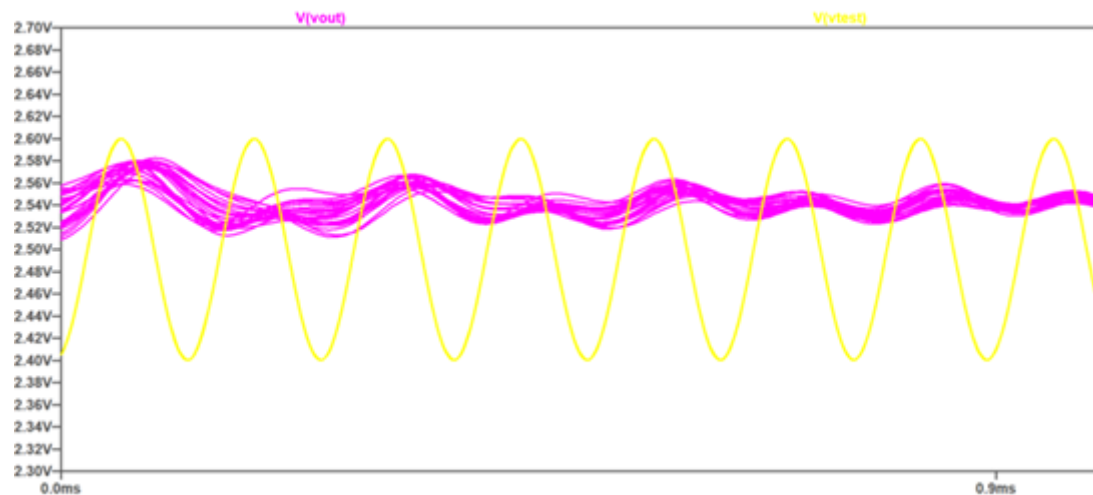


Figure 11 Transient response of the circuit at 7.8kHz

Figure 10 and Figure 11 showcase the time domain response of the circuit, using a 2.5V offset 100mVpp test signal, where the output is parameterized with the 100 random component values. Figure 10 demonstrates that the input signal has a sufficient gain that never clips above the 5V ceiling, maxing out at 4.8V. However, it is worth noting that the minimum amplitude has a significant 1V difference, reaching only 3.8V. While the gain difference seems insignificant in the frequency domain, the difference is significant in the time domain due to the decibels logarithmic scaling. Figure 11 shows the output signal at 7.8kHz, where the signal must be sufficiently attenuated. The figure demonstrates this, where the output signal is at most 10mVpp, which is significantly lower than the input, and especially when compared to the passband signal from Figure 10. The output signal is observed to be largely inconsistent, not having the characteristic sinusoidal wave pattern as the input. Additionally, the signal has a slight offset and is no longer centred at 2.5V. However, even with the distorted phase and slight offset, this will not impact the performance of the recording as the signal is too small to be encoded with the 8-bit sampling size.

5 Experimental Results

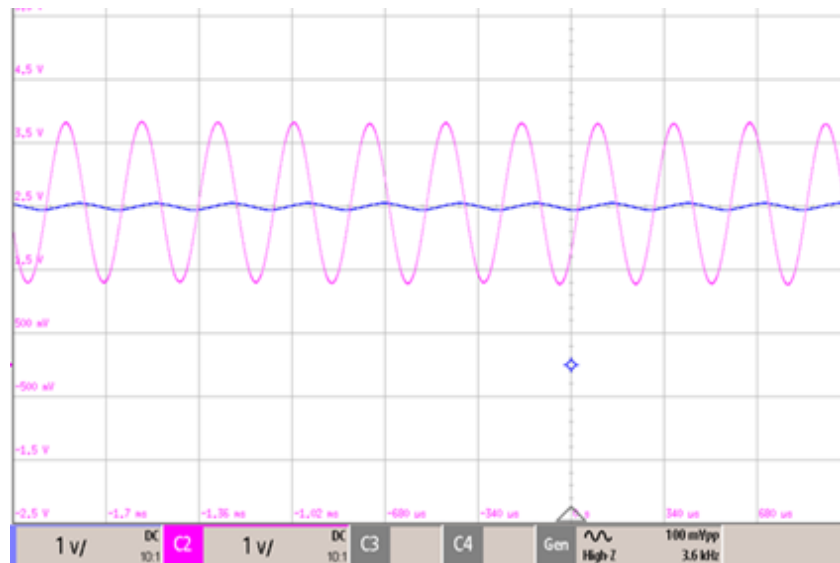


Figure 12 Experimental transient signal for 3.6kHz

Figure 12 shows the oscilloscope probe output of the transient signal for 3.6kHz using a 100mVpp signal generator at the edge of the passband. It shows that the maximum amplitude is close to 3.8V, which falls within the lower bounds as shown in Figure 10 whilst maintaining the same relationship to the input voltage. Therefore, the gain follows the expected results from the simulation, but the components are within the higher ends of the 5% tolerance. So, with higher precision components, it is safe to assume that the gain will reach the desired 4.8V ceiling. The results also demonstrate that the signal has been effectively centred around 2.5V

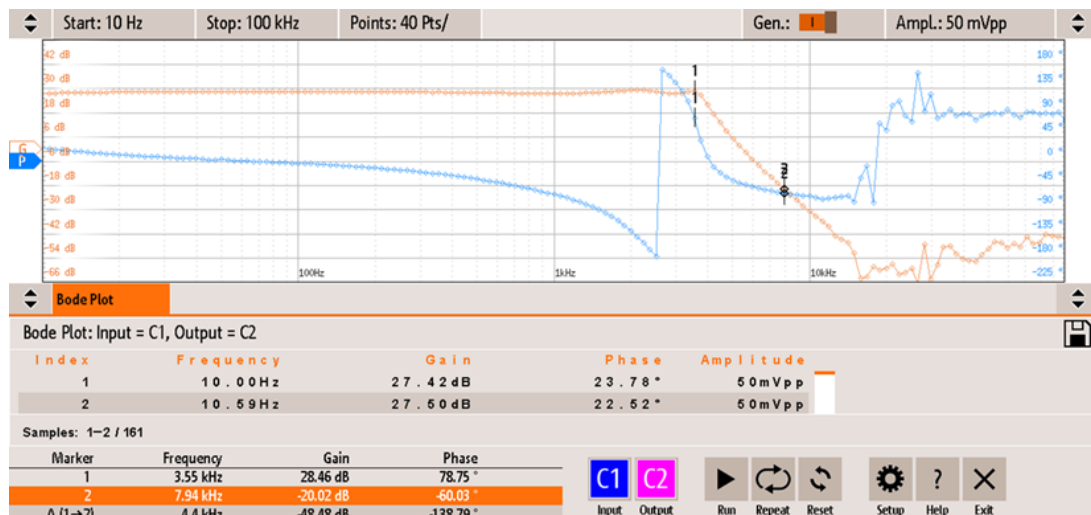


Figure 13 Experimental bode plot

Figure 13 showcases the frequency response of the experimental system. Its shape closely aligns with the theoretical results in Figure 8, showing a similar passband ripple and attenuating at similar points. The high pass filter component is not represented, since the oscilloscope bode plot function does not work for frequencies under 10Hz. The included cursors demonstrate that the circuit has met the design requirements, where the maximum passband frequency is 3.6kHz, above the desired 3.4kHz, and attenuates by -48db at 7.8kHz. While the cursor is 100Hz ahead of 7.8kHz due to the

sampling frequency, interpolation between the previous point against the gain at 500Hz (which has a higher gain than the edge), concludes that the -48db cutoff point is between 7.8 and 7.82kHz. Since the Nyquist rate is exactly 7.8125kHz, it is more likely for the system to fall within the desired range. However, there is a slight possibility that it is slightly over. In the case that it is over by only 50Hz, the aliased signal will be less than 1% power, and so any artifacts or noise from aliasing will be barely audible, especially compared to the innate noise from the low 8-bit quantization level. Following the theoretical bode plot in Figure 9, and the results from the transient response, it is clear that the constructed circuit has components with the top 10% worst tolerances, leading to a higher cutoff frequency and lower overall gain. While this can be remedied by replacing the components, testing on what is close to a manufacturing defect gives a strong argument for the reliability of the design. This is because, if the audio output is clearly audible on the worst component values, then the system is guaranteed to be effective for the full range of component tolerances and is reliable for large scale manufacturing.

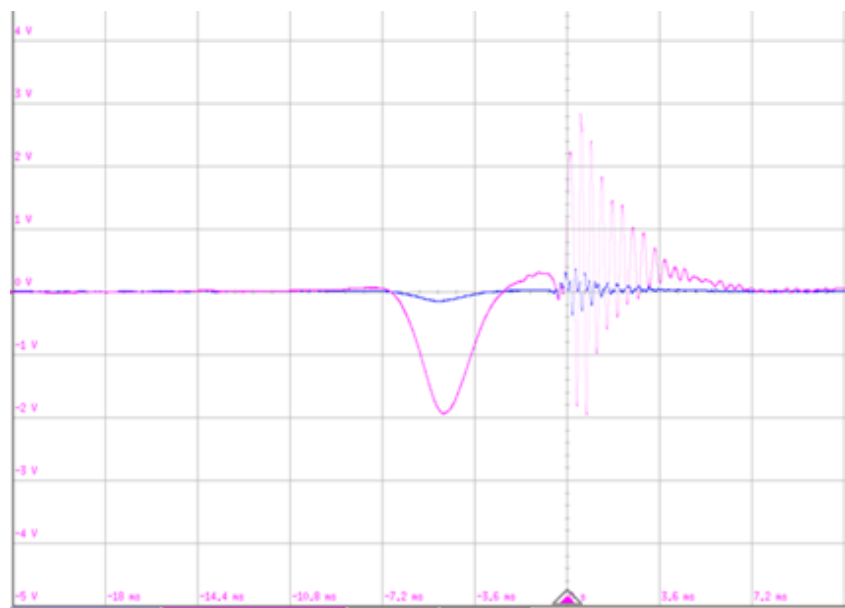


Figure 14 Experimental result from the microphone input.

Figure 14 showcases the output of the microphone, set to AC coupling (which causes the signal to be centred at 0) for the snap of the fingers. A finger snap was chosen to test the microphone instead of speech since it is quick, easy, loud, and lies within 1 to 3kHz, making for a repeatable test [6]. From the waveform, it shows that the signal has been captured with an appropriate gain. The amplitude is shown to reach a peak of 2.8V, which corresponds to 5.3V with the additional 2.5V bias. Since 5.3V is not achievable through the op amps, it suggests misconfigurations in the AC coupling setting of the oscilloscope, and is likely that the finger snap instead almost reaches the voltage ceiling due to its increased sound pressure compared to human speech.

After testing that the experimental results match the theoretical values, the circuit is connected to the ATmega controller to test the recorded signal quality. Playback of the recorded signal sounded slightly noisy and harsh, similar to a voice transmitted over the radio, and yet the spoken words remained clear. The passband ripple did not have much of a noticeable impact on the audio quality. This behaviour matches the expected result, since the passband frequency was designed to be the same as the standard used in radio transmission. It is unknown how much of the noise the speaker contributed, since there is no loaded example of a standard quality 44100 sampling rate 16-bit quality signal to use as a reference.

The final prototype is shown in Figure 15.

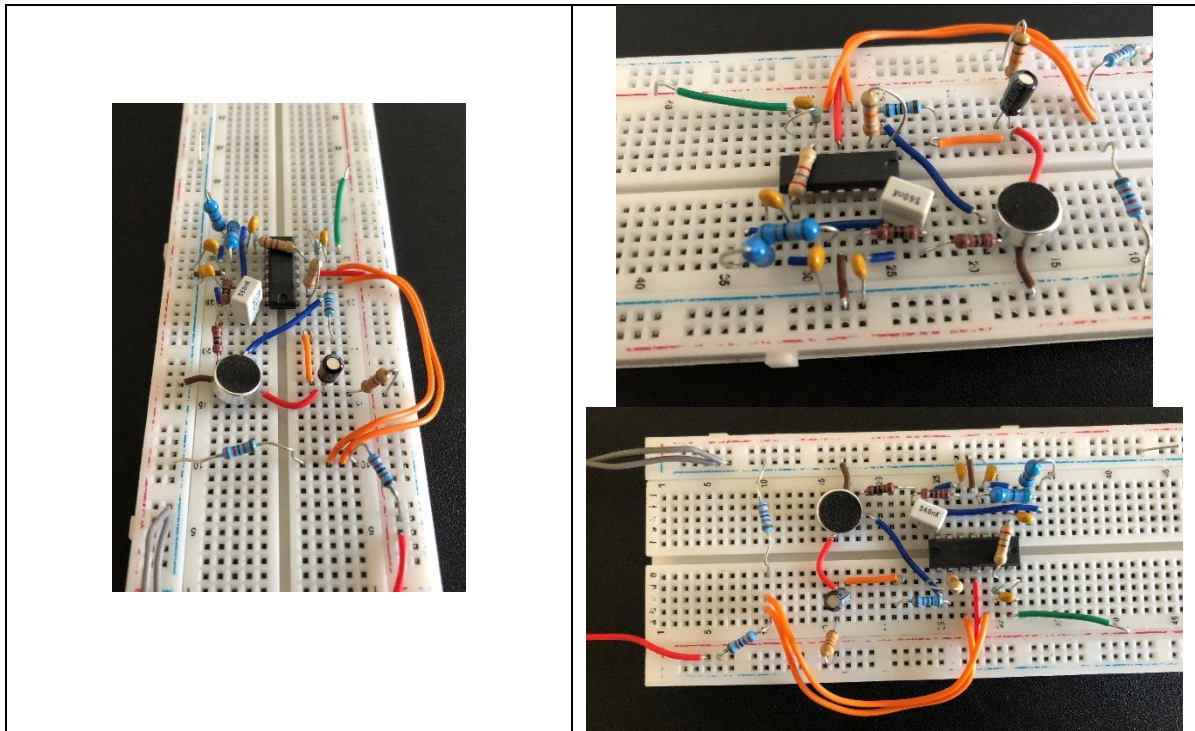


Figure 15 Constructed prototype. The red wire is the input voltage, grey and brown are ground, green is the output, and orange is the reference voltage and input after the microphone interfacing. From the bottom right image, the bottom left section of the IC is the high pass filter, the bottom right is the final filter stage and gain, and the top section is the fourth order low pass filter.

6 Conclusion and Future Improvements

Overall, the circuit design meets all of the required parameters while still being usable in the worst case in manufacturing. The design effectively attenuates the full signal before the Nyquist rate avoiding aliasing, while maintaining a reasonable gain for a louder output recording and centring the signal around 2.5V. The design managed to meet the minimum standard passband used in most audio transmission channels (3.4kHz), while keeping the entire design inside one op amp and without destructive interference.

The main limitation of this design, however, is the slight passband ripple from the Chebyshev filter topology required to minimize costs. While it was barely audible in the recording, a flatter passband would improve the clarity of consonants within the 2.8-3.6kHz range. To improve the output quality, the Chebyshev filter can be replaced by Butterworth filters at the cost of using more op amps, to flatten the passband gain. This, however, would increase the cost of the circuit as additional ICs are required.

Some of the observed noise or variance in the signal may also be caused by the input currents to the op amp. The design assumes the inputs draw no current, but practically they draw a small current. This impacts the voltage reference, since this additional current loads the voltage divider, making the reference voltage unstable. So, an improvement to the design would be to use one op amp from the added IC to buffer reference voltage. This buffer also allows for the filter stage topology to have a gain, offloading some of the gain from the final stage of the circuit to lower the risk of damage. An alternative that keeps the same design is to construct one voltage divider per section rather than sharing, and increasing the impedance to lower the impact of the current draw.

Improvements to the recording can also be made without modifications to the circuit, by making use of digital audio processing. The variance in the passband gain is the main limiting factor to audio quality, outside of the passband frequency. Since there is no aliasing in the controller's recorded signal, it can also perform some slight equalization to the signal to minimize the impacts of the passband ripple, making use of the Fast Fourier Transform function (FFT). While microcontrollers are memory constrained, FFT is highly efficient and does not allocate much additional memory, and so would be viable for the ATmega's specifications.

7 References

- [1] WatElectronics, "Nyquist Criteria : Theory, Calculations, Applications, Sample Problems," WatElectronics, 7 February 2024. [Online]. Available: <https://www.watelectronics.com/nyquist-criteria/>. [Accessed 4 May 2026].
- [2] L. Cope, "What Is Passband Ripple and How Does It Affect Signal Quality?," Engineering Fix, 2 November 2025. [Online]. Available: <https://engineerfix.com/what-is-passband-ripple-and-how-does-it-affect-signal-quality/>. [Accessed 7 May 2026].
- [3] International Telecommunication Union, "ITU-T Recommendation P.310: Transmission characteristics for telephone band (300–3400 Hz) digital telephones," International Telecommunication Union (ITU), Geneva, 2003.
- [4] N. R. French and J. C. Steinberg, "Factors Governing the Intelligibility of Speech Sounds," *The Journal of the Acoustical Society of America*, vol. 19, no. 1, pp. 90-119, 1947.
- [5] Texas Instruments, "Analog Engineer's Circuit Amplifiers Low-Pass, Filtered, Inverting Amplifier Circuit," January 2021. [Online]. Available: https://www.ti.com/lit/an/sboa293a/sboa293a.pdf?ts=1777512248153&ref_url=https%253A%252F%252Fwww.bing.com%252F. [Accessed 9 May 2026].
- [6] L. Torres, "The Science Behind Snapping: How Fingers Create That Distinctive Sound," 12 October 2025. [Online]. Available: <https://soundcy.com/article/how-does-snapping-make-sound>. [Accessed 13 May 2026].

Appendix A – MATLAB Code

```
%% Filter specs
fsamp = 15.625e3;

fs = fsamp/2; % stopband frequency, Hz
fp = 3.6e3; % passband frequency, Hz

ws = 2*pi*fs;
wp = 2*pi*fp;

Amin = 20*log10(1/2^8); % stopband attenuation, dB 48dB
Amax = 1.5; % passband attenuation, dB

[n_cheb, wn_cheb] = cheb1ord(wp, ws, Amax, Amin, 's');

%% plot ideal cheby filter
[b a] = cheby1(n_cheb, Amax, wn_cheb, 'low', 's');
H = tf(b,a);
figure(1);

h = bodeplot(H); % display bode plot
setoptions(h,'FreqUnits','Hz'); % change units to Hz
setoptions(h,'FreqUnits','Hz','PhaseVisible','off'); % remove phase plot
grid on;

[z, p, k] = tf2zpk(b, a);

%%%%%%%% stage A %%%%%%%%%
a1A = -p(1)-p(2);
a2A = p(1)*p(2);

wnA = sqrt(a2A);
QA = wnA/a1A;

fprintf(1,'Stage A: wn = %g rad/s, Q = %g\n',wnA,QA);

%%%%%%%% stage B %%%%%%%%%
a1B = -p(3)-p(4);
a2B = p(3)*p(4);

wnB = sqrt(a2B);
QB = wnB/a1B;

fprintf(1,'Stage B: wn = %g rad/s, Q = %g\n',wnB,QB);

%%% stage C %%%
wnC = abs(real(p(5)));

%%%%%%%% stage A design %%%%%%%%%
KA = 1;

nA = (2*QA)^2;

C1A = 3.3e-9;
C2A = nA*C1A;

R1A = 1/(wnA * C1A * sqrt(nA));
```

```

R2A = R1A;

fprintf(1,'R1A = %g, R2A = %g, C1A = %g, C2A = %g\n',R1A,R2A,C1A,C2A);

%%% preferred values
R1A = 2.2e3;
R2A = R1A;
C1A = 3.3e-9;
C2A = 220e-9;

%%%%%%%% stage B design %%%%%%%%%%
KA = 1;
nB = (2*QB)^2;
C1B = 3.3e-9;
C2B = nB*C1B;

R1B = 1/(wnB * C1B * sqrt(nB));
R2B = R1B;

fprintf(1,'R1B = %g, R2B = %g, C1B = %g, C2B = %g\n',R1B,R2B,C1B,C2B);

%%% preferred values
R1B = 20e3;
R2B = R1B;
C1B = 3.3e-9;
C2B = 10e-9;

%%%%%%%% Passive RC + inverting gain stage %%%%%%%%%%

C1P = 10e-9;    % example values
R2P = 1 / (wnC * C1P);
KC = -4;
R1P = -R2P / KC;

KC = -R2P/R1P;

fprintf(1,'R1P = %g, R2P = %g, C1P = %g\n',R1P,R2P,C1P);

```

Appending B – Ltspice Netlist

```
* Generated by LTspice 26.0.1 for Windows.
Ggain 0 Vmic N003 0 10u
Ibias Vmic 0 0.2m
R1 Vin Vmic 12k
Vvoice N003 0 SINE(0 1 2k) AC 1
Vsupply Vin 0 5V
X$U1 VRef N002 Vin 0 Vmc TL974 ;$pnba In+)In-)V+)V-)OUT
R2A Vmc N002 {R2A} tol=10
C1A P001 Vtest {C1A}
R1A N002 P001 {R1A} tol=10
X$U2 N007 N005 Vin 0 N005 TL974 ;$pnba In+)In-)V+)V-)OUT
R2B N007 N004 {R2B} tol=10
C2B N005 N004 {C2B}
C1B N007 0 {C1B}
V1 Vtest 0 SINE(2.5 100m 7.8k) AC 1
R8 Vref 0 22k
R9 Vin Vref 22k
X$U3 N008 Vlp Vin 0 Vlp TL974 ;$pnba In+)In-)V+)V-)OUT
R1C N006 N005 {R1C} tol=10
R2C N008 N006 {R2C} tol=10
C2C Vlp N006 {C2C}
C1C N008 0 {C1C}
X$U4 Vref N001 Vin 0 Vout TL974 ;$pnba In+)In-)V+)V-)OUT
R1B N004 Vmc {R1B} tol=10
R2D Vout N001 {R2D} tol=10
R1D N001 Vlp {R1D} tol=10
C1D Vout N001 {C1D}
* .tran 0 10m 1m
* Audio input where 1 Vp represents \n100% normal speaking volume
* Electret mic.
.include TL974.sub
.ac dec 100 1 1Meg
* Voltage net lines
* Reference voltage
* Test voltage (to bypass \nmicrophone in AC analysis)
* Microphone connection (5hz high pass filter)
* DC input voltage
* Small amplification and low pass filter (fc~900)
* Unity gain sallen key LP filter stage 1 (fc~5.7k)
* Unity gain sallen key LP filter stage 2 (fc~2.7k)
* Vout: fc=3.6kHz, -48db at 7.8kHz
.param R1A=mc(10k, 0.05) R2A=mc(39K, 0.05) C1A=mc(3.3u, 0.05)
.param R1B=mc(1k,0.05) R2B=mc(1k,0.05) C1B=mc(3.3n,0.05) C2B=mc(560n,0.05)
.param R1C=mc(6.8k,0.05) R2C=mc(6.8k,0.05) C1C=mc(3.3n,0.05)
C2C=mc(33n,0.05)
.param R1D=mc(2.7k,0.05) R2D=mc(18k,0.05) C1D=mc(10n,0.05)
.MEAS AC mag_pb MAX db(V(vout)) FROM=100 TO=2k
.MEAS AC f_48 WHEN db(V(vout))=mag_pb-48
.step PARAM AC 1 100 1
.backanno
.end
```

Appendix C – Experimental Results

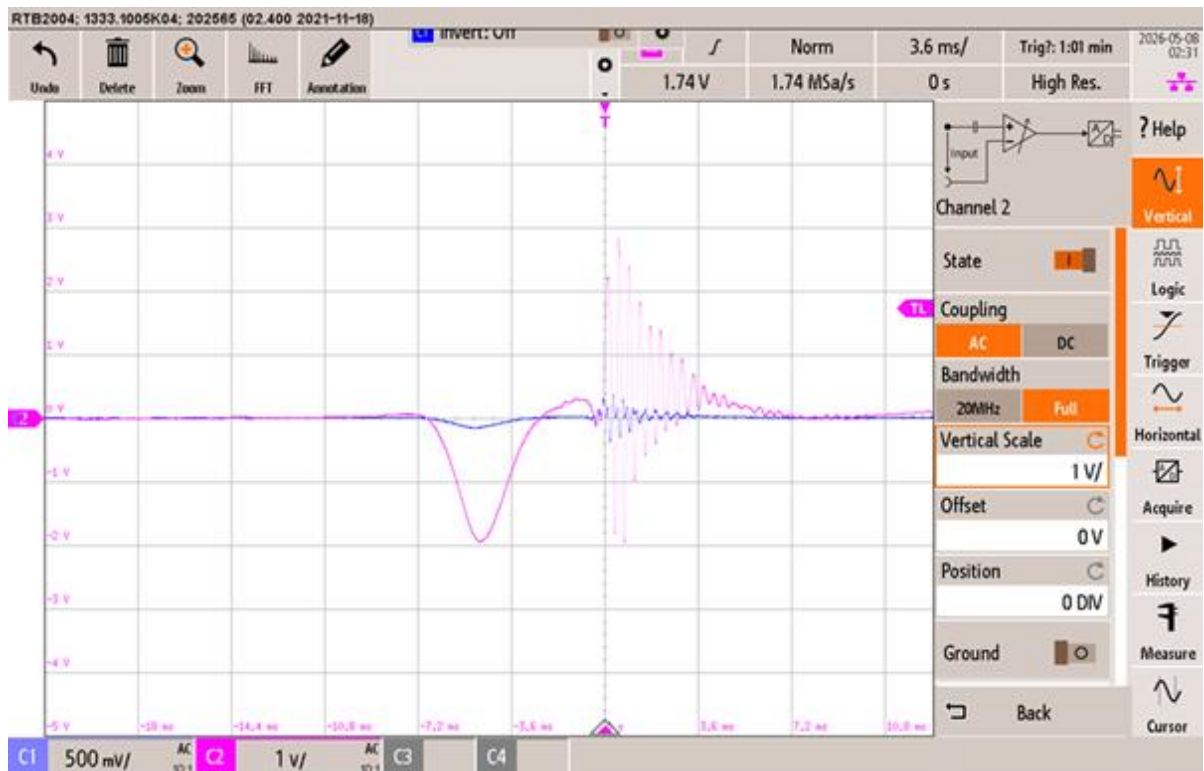


Figure 16 Raw oscilloscope output of a repeating finger snap.



Figure 17 Raw oscilloscope bode plot.

The bode plot shows the frequency before the attenuation is at 3.6kHz, and the frequency near 7.8kHz has a gain of -48dB, matching the theoretical results. The noise after 10kHz does not impact the voice recording, since it is too small to be sampled in 8 bits.



Figure 18 Raw bode plot with differing measurement points, allowing interpolation between the two bode plots.

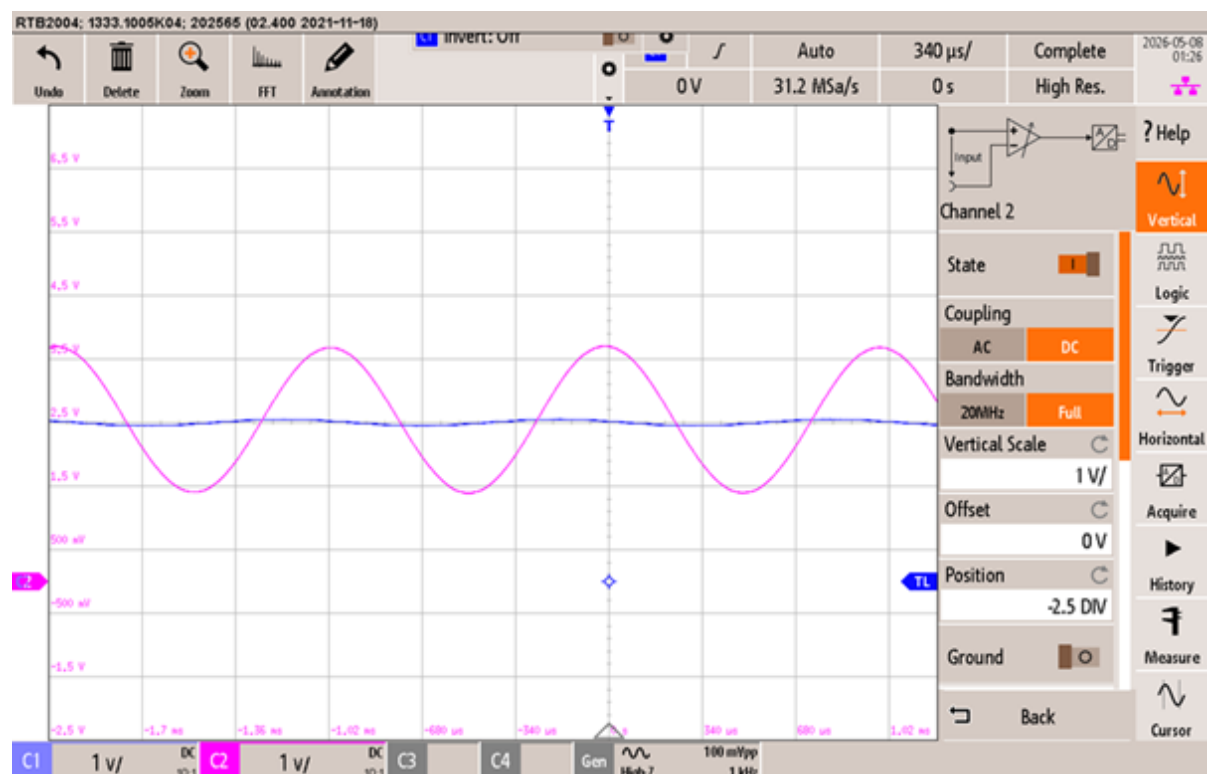


Figure 19 Raw oscilloscope output of a 1kHz frequency. C1 and C2 are the input and output respectively.

This demonstrates that the gain has not caused clipping, as seen by the clean sinusoidal signal.

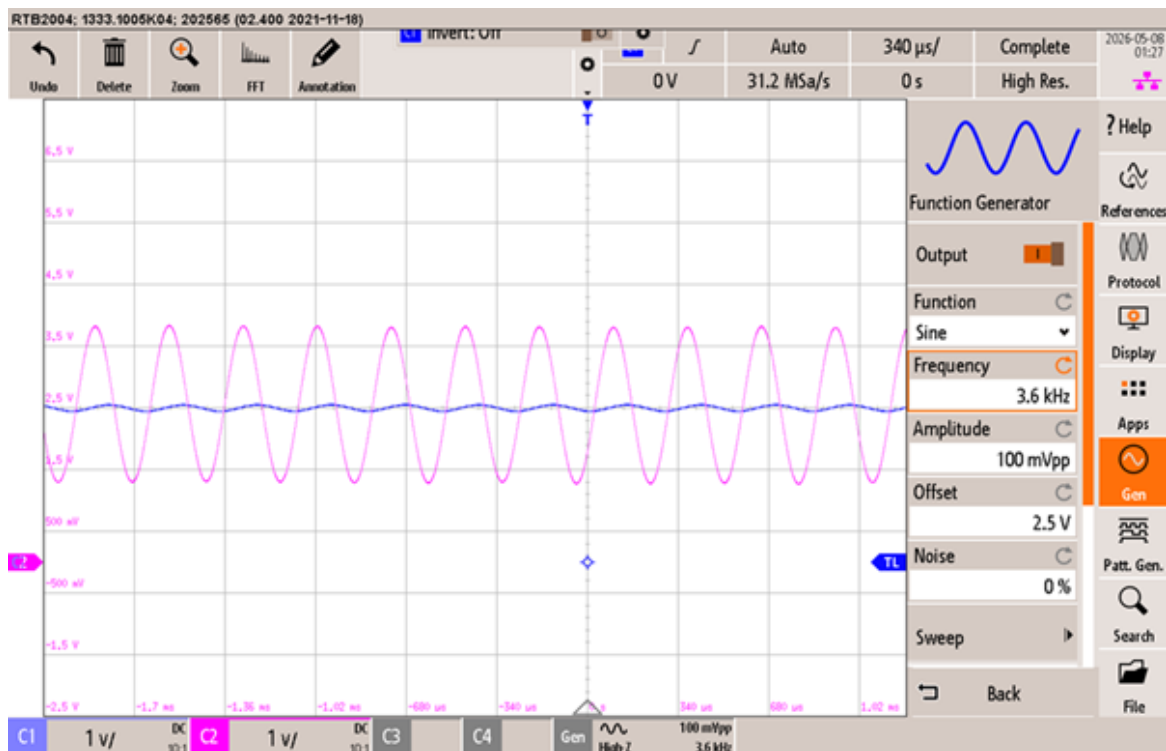


Figure 20 Raw oscilloscope output at the edge of the passband (3.6kHz). C1 and C2 are the input and output respectively.

This probe output shows that at the edge of the passband, it still maintains the same voltage peak-to-peak as in the middle of the passband (1kHz).

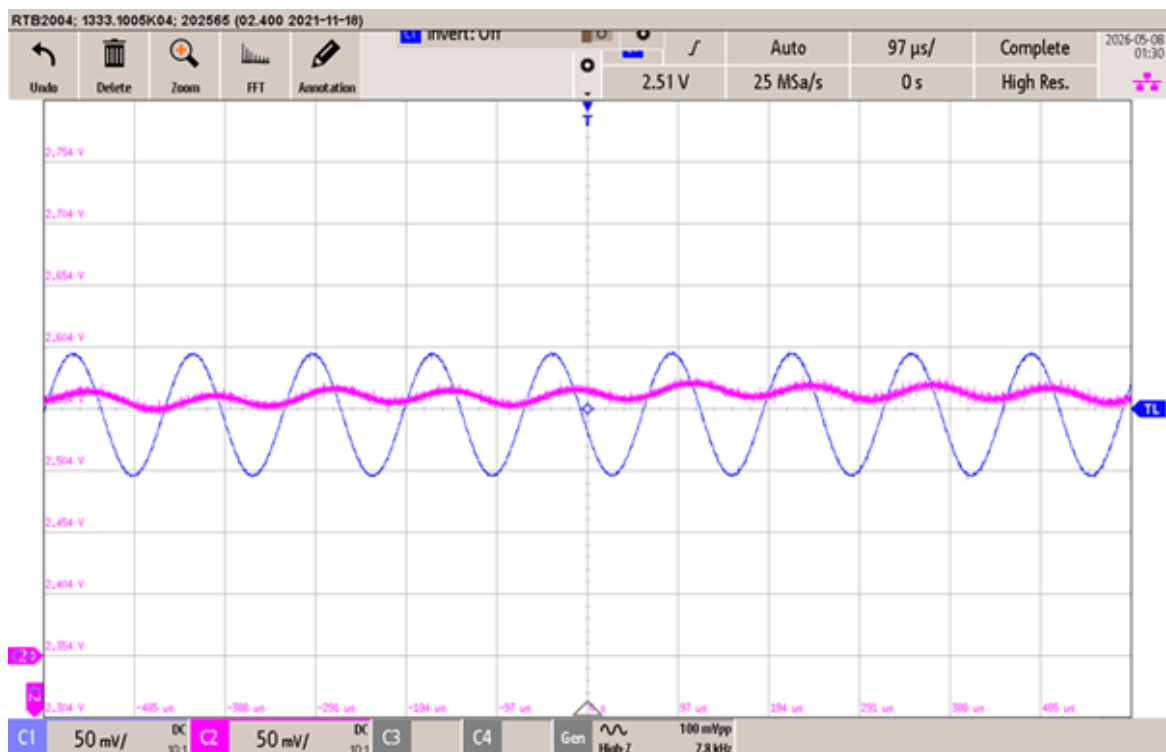


Figure 21 Oscilloscope output when fully attenuated (7.8kHz). C1 and C2 are the input and output respectively

Note that for this output, the scale is 50mV compared to the 1V of the previous scale, so while the output is still large relative to the input, it is significantly attenuated compared to the passband.

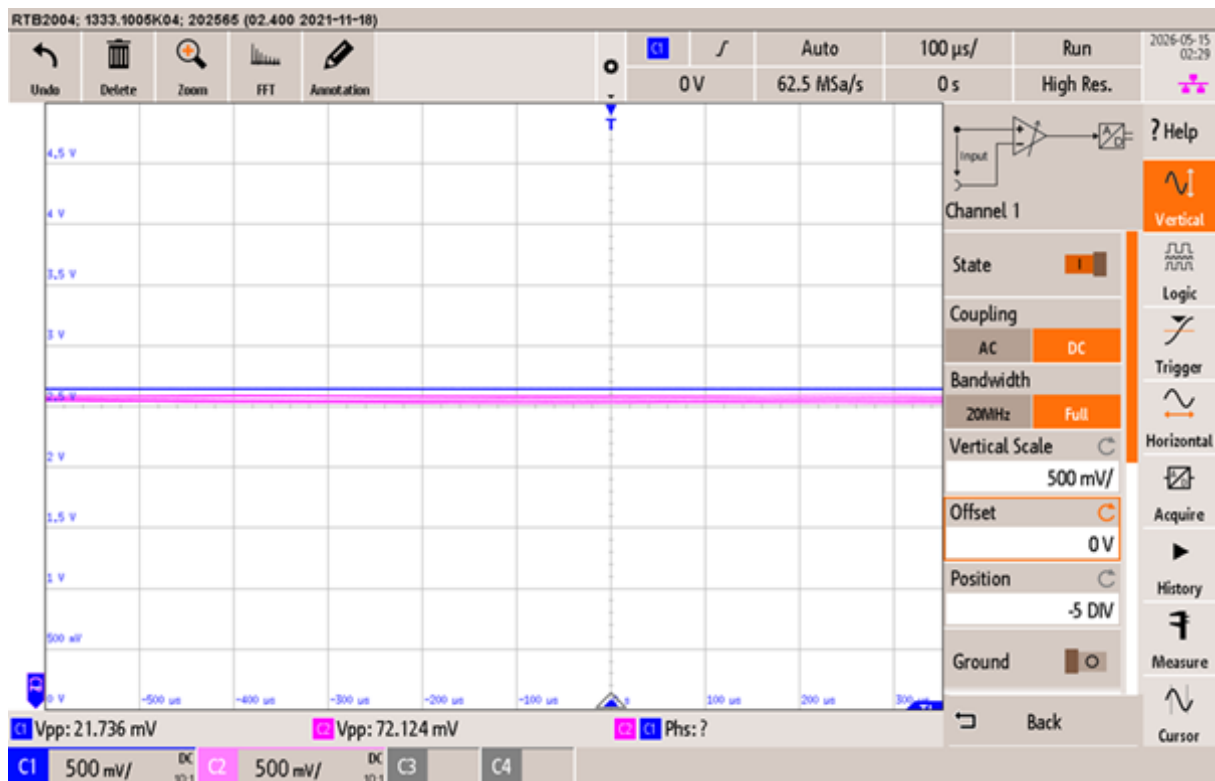


Figure 22 Input output relationship for C1 and C2 respectively, showing the DC bias offset

This probe output shows that the DC offset of the microphone signal has been mostly centred using the pulldown resistor. Because of the component tolerances, it only reaches 2.6V instead of 2.5V, but this is negligible as the microphone buffer circuit fixes this as shown by the channel two output.